MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6211

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of MIL-E-1.

DESCRIPTION: Medium-Mu Twin Triode for "On-Off" Applications

Outline --- 6-2 (EIA)

Base --- E9-1

Envelope --- T6-1/2

Cathode --- Coated unipotential

Base connections:

Pin No.:	1	2	3	4	5	6	7	8	9
Element:	2p	2g	2k	h	h	1p	1g	1k	hct

ABSOLUTE MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec	Pp/p	Ik/k	Ehk	Ig/g	Rg/g	T Envelope
Unit:	Vdc	Vdc	Vdc	W	mAdc	Vdc	mAdc	meg	°C
Maximum:	13.2	200	+1.0	1.5	16	±180	2.0	0.1	120
Minimum:	12.0	---	-100	---	--		---	---	---
TEST CONDITIONS:	12.6	150	0	---	--	0	---	0	---

GENERAL:

Qualification - Required

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Method	Requirement or test	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 1							
1266	Total grid current	Ec=-3.0V, Rg=0.5 meg, (see note 1)	0.65	II	Ic:	---	-1.0	μAdc
1256	Electrode current (1) (anode)	Eb=85V, Rg=0.5 meg, returned to +100 Vdc, (see note 1)	0.65	II	Ib:	12.0	20.0	mAdc
1256	Electrode current (2) (anode)	Eb=100 Vdc, Ec=-2 Vdc; (see note 1)	0.65	II	Ib:	3.5	---	mAdc
----	Grid voltage	Ec/Ib=100 μA, (see note 1)	0.65	II	Ec:	---	-10.0	Vdc
1201	Shorts and discontinuity detection		0.4	II	--	---	----	----
	Quality conformance inspection, part 2							
1211	Insulation of electrodes	(see note 1) g-all p-all	2.5	S-3	R: R:	100 100	---- ----	meg meg
1256	Electrode current (3) (anode)	Ef=11.5 V, Ec=-2 Vdc, Eb=100 Vdc, (see note 1)	6.5	Code E	Ib:	3.0	----	mAdc
1306	Transconductance	Eb=100 Vdc, Rk=470 ohms;(see note 1) Ck=1000μf	6.5	Code E	Sm:	2700	4500	μmhos
1336	Heater-cathode leakage	Ehk=+100 Vdc Ehk=-100 Vdc	1.5	I	Ihk: Ihk:	---- ----	20 20	μAdc μAdc
1301	Heater current		1.5	I	If:	138	162	ma
------	Grid voltage difference between units:		6.5	Code E	ΔEc:		1.5	Vdc
----	Intermittent shorts	(See notes 2, 3, and 16)	1.5	I	----	---	---	---
1331	Direct-interelectrode capacitance	No shield	---	---	Cg-p: Cin: Cout1: Cout2: C1p-2p: C1a-2a:	1.84 2.30 0.28 0.25 0.27 ----	2.60 3.40 0.80 0.66 0.85 0.06	μμf μμf μμf μμf μμf μμf

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Method	Requirement or test	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits		Unit
						Min	Max	
	Quality conformance inspection, part 2 (Continued)							
2126	Glass strain		4.0	I	----	---	---	---
1105	Permanence of marking		---	---	----	---	---	---

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Method	Requirement or test	Conditions	AQL (percent defective)	Insp. level or code	Allowable defectives per characteristics		Symbol	Limits		Units
					1st sample	Combined samples		Min	Max	
	Quality conformance inspection, part 3 (see note 11)									
1506	Heater cycling life test:	Ef=15.0 V, Ec=Eb=0, Ehk=+180 V, 1 min. on, 4 min. off note 8	---	----	-	-	----	---	----	-----
----	Stability life test: (2 hours)	Intermittent life test conditions (Ec=0)		Code H						
----	Stability life test end points:	Electrode current (1) (anode)	1.0				Ib:	9.6		mAdc
		Electrode current (2) (anode)	1.0				Ib:	2.8		mAdc
----	Survival rate life test: (100 hours)	Intermittent life test conditions (Ec=0) or equivalent (see note 12)	---	Code H	-	-	----	---	----	-----
----	Survival rate life test end points:	Shorts and continuity	0.65	----	-	-	----	---	----	-----
----	Intermittent life test:	Ef=12.6 V, Ebb=150 V, Ehk=180V TA=room, (see notes 4, 11, and 13)								
----	Intermittent life test end points: 1000 hours	(see notes 6, 10 and 9)								
		Interface resistance (see note 5)	----	-----	2	4	Ri:	---	25	ohms
		Intermittent shorts	----	-----	2	4	--	---	--	-----
		Shorts and continuity	----	-----	1	2	--	---	---	-----
		Total grid current	----	-----	1	2	Ic:	---	-1.5	μAdc
		Electrode current (1) (anode)			2	4	Ib:	10.8	21.0	mAdc
		Electrode current (2) (anode)			2	4	Ib:	3.0	----	mAdc

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Method	Requirement or test	Conditions	AQL (percent defective)	Insp. level or code	Allowable defectives per characteristics		Symbol	Limits		Units
					1st sample	Combined samples		Min	Max	
	Quality conformance inspection, part 3 (see note 11) (Continued)									
----	Intermittent life test end points: 1000 hours	Heater cathode leakage	----	-----	2	4	+Ihk: -Ihk:	--- ---	30 30	µAdc µAdc
		Insulation of electrodes	----	-----	2	4	R: R:	50 50	-- --	meg meg
		Electrode current (1) (anode) Avg. change: (see note 8)	----	-----	-	-	ΔIb: t	---	15	%
		Total defectives	----	-----	5	9	--	---	--	----
----	Intermittent life test end points: 500 hours	Electrode current (1) (anode)	----	-----	2	4	Ib:	9.6	--	mAdc
		Electrode current (2) (anode)			2	4	Ib:	2.8	--	mAdc
		Interface resistance (see note 5)			2	4	Ri:	---	25	ohms

NOTES:

1. Each unit shall be tested separately. Unit not under test shall be grounded.

2. Tube shall be tested on d.c. Thyratron Shorts Test Set, or equivalent. Minimum sensitivity shall be as follows:

Resistance	Time
1000 ohms	2 microseconds
100,000 ohms	15 microseconds
2.5 meg	1 millisecond

3. Each tube shall be tapped six times, three times in each of two planes using felt tapper consisting of felt mallet head 1.0 inch diameter and 1.0 inch long with 1/4 inch concentric hole securely glued flush with end of an 8 by 1/4 inch plastic rod. Weight of felt shall be approximately 3.0 grams.

4. Sample size shall be 20 tubes per lot. Of these, ten shall operate with Ecc=-25 Vdc, Rg/g=47K, Rb/p=4. OK and Ehk=-100 Vdc. The other ten shall operate with Ecc=+150 Vdc, Rg/g=1.8 meg, Rb/p=4.3K and Ehk=+180 Vdc.

5. Interface resistance shall be read on Waters Cathode Interface Impedance Bridge Type 1001-C, or equivalent. Test conditions: Eb=75 Vdc, Ib=2.0 mAdc, Ef=11.4. Each unit shall be tested separately. Read at 500 and 1000 hours only.

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NOTES: (Continued)

6. Tests shall be made in the order listed. Combined sample size equals 40 tubes, that is, 20 first sample, 20 repeat.

7. The no-load to steady state full load regulation of the heater voltage supply shall be not more than 3.0 percent. This test shall be made on a lot by lot basis.

$$\Delta Ib = \frac{\text{Avg. anode current (1) at t = 0 - Avg. anode current (1) at t = 1000}}{\text{Avg. anode current (1) at t = 0}}$$

The 15 percent Ib avg. limit applies to downward shift in anode current only.

9. No lot shall be released for shipment until life test sample for that lot passes 1000 hour life test.

10. Order for evaluation of life test defects.
If a tube is defective for more than one attribute characteristic, the characteristic appearing first in the Life test end points shall constitute the failure.

11. Destructive tests.
Tubes subjected to the following destructive tests will not be accepted under this specification:

Heater-cycling life test.
Intermittent life test.

12. For Survival rate life test, the equivalent conditions shall be interpreted as having the same heater voltage (Ef) and heater-cathode voltage (Ehk) as the Intermittent life test; and the same interruptions of method 1501 of MIL-STD-1311, as the Intermittent life test. The electrode voltages shall be such that the element dissipations are not less than 80 percent, nor more than 100 percent of Intermittent life test plate dissipation. These voltages shall be maintained within the limits of plus 200, minus 50 percent of of the Intermittent life test voltages.

13. The Intermittent life test sample shall be read at the following times:

0 hours
500 hours (plus 48 hours; minus 24 hours)
1000 hours (plus 48 hours; minus 24 hours)

14. This value is for operation under fixed bias conditions with cathode bias, Rg may be 0.5 megohm maximum.

15. Test with units connected in parallel.

16. After single indication of short, a tube shall be given three additional cycles of six taps each. A second indication for any type of high resistance short constitutes a failure.

Preparing activity:
Navy - EC

(Project 5960-N875)

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NAVAL ELECTRONIC SYSTEMS COMMAND
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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